🧑‍🏫 Assignment 1: Student Management System (Basic CRUD)

🎯 Objective

Understand basic **ADO.NET CRUD operations using SqlConnection, SqlCommand**

📌 Scenario

You are building a **Student Management Module** for a school.

🗃️ Database Table

```
CREATE TABLE Students (
    Id INT PRIMARY KEY IDENTITY,
    Name NVARCHAR(100),
    Age INT,
    Grade NVARCHAR(10)
);
```

✅ Tasks

1. **Create (Insert Student)**
    - Add a new student record using INSERT query
2. **Read (Get All Students)**
    - Fetch all records using SqlDataReader
3. **Update (Modify Student)**
    - Update student Grade by Id
4. **Delete (Remove Student)**
    - Delete student using Id

🧠 Concepts to Apply

- SqlConnection

- SqlCommand
- ExecuteNonQuery()
- ExecuteReader()
- Connection string handling

🧑‍🏫 Assignment 2: Employee Management with Stored Procedures

🎯 Objective

Learn **Stored Procedures + Parameterized Queries**

📌 Scenario

You are developing an **HR system**.

💾 Table

```
CREATE TABLE Employees (
    EmpId INT PRIMARY KEY IDENTITY,
    Name NVARCHAR(100),
    Salary DECIMAL(10,2),
    Department NVARCHAR(50)
);
```

📜 Stored Procedure Example

```
CREATE PROCEDURE InsertEmployee
    @Name NVARCHAR(100),
    @Salary DECIMAL(10,2),
    @Department NVARCHAR(50)
AS
BEGIN
    INSERT INTO Employees VALUES (@Name, @Salary, @Department)
END
```

✅ Tasks

1. Insert employee using **Stored Procedure**
2. Fetch employees by Department
3. Update Salary using Stored Procedure
4. Delete employee using parameterized query

🧠 Concepts to Apply

- CommandType.StoredProcedure
- SqlParameter
- Avoid SQL Injection
- ExecuteScalar() (optional for counts)

🧑‍🏫 Assignment 3: Product Inventory System (Disconnected Architecture)

🎯 Objective

Work with **DataSet, DataAdapter (Disconnected Model)**

📌 Scenario

You are building an **Inventory System** for a retail shop.

🗃️ Table

```
CREATE TABLE Products (
    ProductId INT PRIMARY KEY IDENTITY,
    ProductName NVARCHAR(100),
    Price DECIMAL(10,2),
    Stock INT
);
```

✅ Tasks

1. Load data into **DataSet**
2. Display data using **DataTable**
3. Add new product (offline)
4. Update product price (offline)
5. Delete product (offline)
6. Push changes back using SqlDataAdapter.Update()

🧠 Concepts to Apply

- SqlDataAdapter
- DataSet & DataTable
- Disconnected Architecture

👩‍🏫 Assignment 4: Order Management System (Advanced CRUD + Transactions)

🎯 Objective

Implement **Transactions and Error Handling**

📌 Scenario

You are developing an **E-commerce Order System**

🗃️ Tables

```
CREATE TABLE Orders (
   OrderId INT PRIMARY KEY IDENTITY,
   CustomerName NVARCHAR(100),
   TotalAmount DECIMAL(10,2)
);
```

```
CREATE TABLE OrderItems (
    ItemId INT PRIMARY KEY IDENTITY,
    OrderId INT,
    ProductName NVARCHAR(100),
    Quantity INT
);
```

✅ **Tasks**

1. Insert into **Orders**
2. Insert multiple **OrderItems**

👩‍🏫 **Assignment 5: Mini Project – Library Management System**

🎯 **Objective**

Combine all CRUD operations into a **real-world mini project**

📌 **Features**

1. Add Book
2. View Books
3. Update Book Details
4. Delete Book
5. Search Book by Name

🗃️ **Table**

```
CREATE TABLE Books (
    BookId INT PRIMARY KEY IDENTITY,
    Title NVARCHAR(200),
```

```
    Author NVARCHAR(100),
    Price DECIMAL(10,2)
);
```